

CO2-AT1

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1. Genetic Algorithm Execution

Step 1: Fitness Evaluation

The objective function is $f(x) = x^2$. Evaluating each individual in the initial population $\{2, 4, 6, 8\}$:

Step 2: Selection

Individual (x)	Binary	Fitness $f(x)=x^2$
2	0010	4
4	0100	16
6	0110	36
8	1000	64

In genetic algorithms, candidates with higher fitness scores have a higher probability of being selected. The top two candidates with the highest fitness scores are:

- **Parent 1:** \$8\$ (1000) — Fitness = \$64\$
- **Parent 2:** \$6\$ (0110) — Fitness = \$36\$

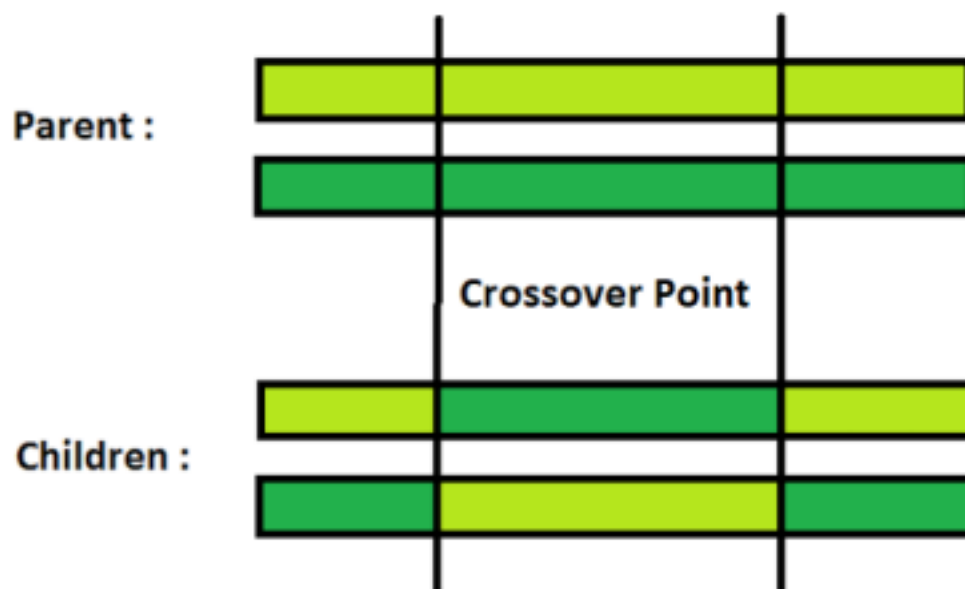
Step 3: Crossover Operation

We perform a single-point crossover at the Least Significant Bit (LSB) position (swapping the last rightmost bit between Parent 1 and Parent 2):

- **Parent 1 (8):** 1 0 0 | 0
- **Parent 2 (6):** 0 1 1 | 0

Exchanging the LSBs:

- **Offspring 1:** 1 0 0 | 0 {1000_2 = 8}\$
- **Offspring 2:** 0 1 1 | 0 {0110_2 = 6}\$

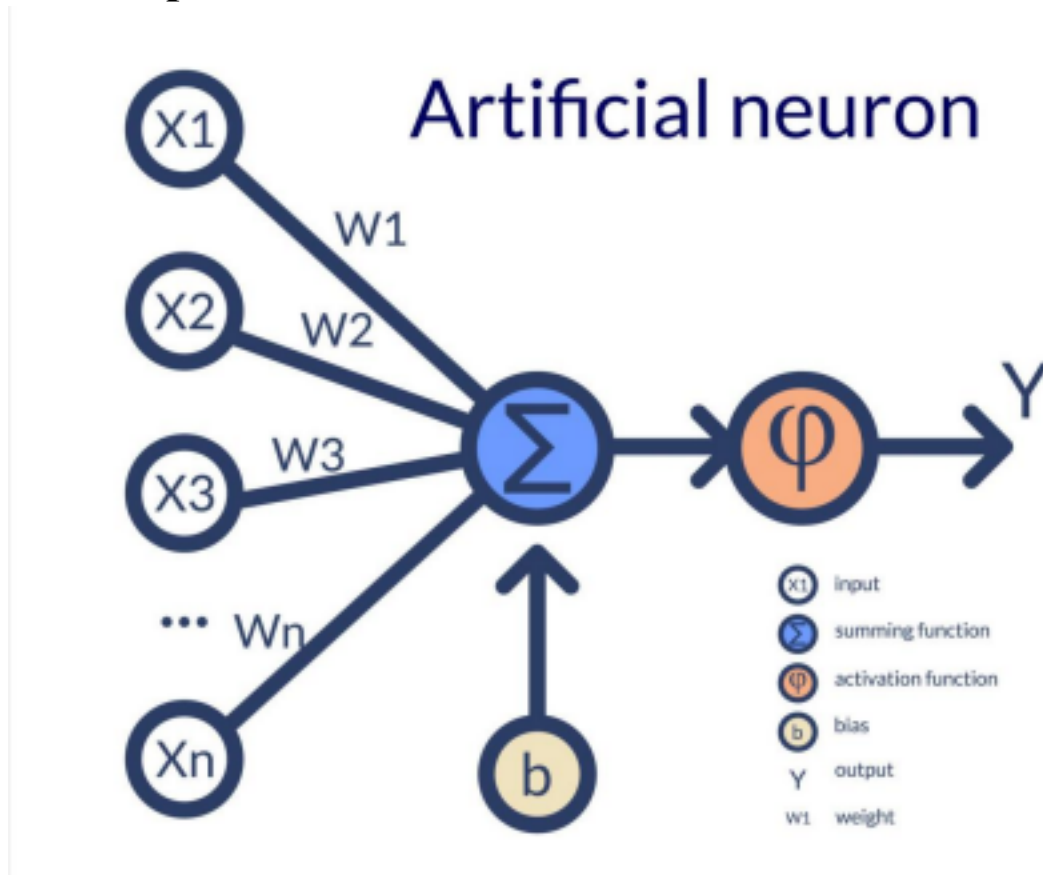


Role of Mutation in Genetic Algorithms

While **crossover** combines existing genetic traits, **mutation** randomly flips one or more bits in a chromosome with a small probability (e.g., \$1\%\$).

- **Prevents Premature Convergence:** It stops the algorithm from getting stuck in local optima by introducing entirely new search paths.
- **Maintains Diversity:** It restores genetic material lost during selection phases, ensuring the algorithm explores unexplored areas of the search space.

2. Perceptron Classification



Given Data:

- Feature 1 (x_1 - Advertisement Exposure): 2
- Feature 2 (x_2 - Income Level): 1
- Weight 1 (w_1): 0.5
- Weight 2 (w_2): 0.3
- Bias (b): -0.4

Step 1: Net Input Calculation (z)

The net input formula is:

$$z = (w_1 \cdot x_1) + (w_2 \cdot x_2) + b$$

Substituting the values:

$$\begin{aligned} z &= (0.5 \cdot 2) + (0.3 \cdot 1) + (-0.4) \\ z &= 1.0 + 0.3 - 0.4 = \mathbf{0.9} \end{aligned}$$

Step 2: Step Activation Function

Applying the standard unit step function $f(z)$:

$f(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$
Since $z = 0.9 \geq 0$:

$$f(0.9) = \mathbf{1}$$

Conclusion

The perceptron outputs **1**, predicting that **the customer will make a purchase**.